William Chang

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EDUCATION

University of California, Berkeley GPA: 3.96, December 2026

B.A. Data Science w/ domain emphasis in Economics

Relevant Coursework: Machine Learning, Data Engineering, Data Mining and Analytics, Data Structures and Algorithms, Principles and Techniques of Data Science, Probability, Linear Algebra, Advanced Programming in R, Data Science for Economists, Econometrics

WORK EXPERIENCE

Meta Menlo Park, CA | May 2026 – August 2026

Data Scientist Intern

- **Drove +0.015% Instagram sessions growth for high-priority countries** by targeting personalized ML ranking models based on user-level responsiveness, capturing a **2.5× higher engagement uplift** while **saving $5M+ in annual capacity spend** compared to a full rollout
- **Analyzed A/B tests across 300M+ exposed users** by training **PySpark Causal Forest models** to predict Heterogeneous Treatment Effects, developing a **Claude Code skill** that automated post-experiment user targeting to **reduce manual modeling time by 90%+**
- **Identified $7.5M in annual server capacity savings** by evaluating revenue against infrastructure costs across global user segments, strategically targeting capacity scale-downs for recommendation systems while **preserving revenue and key engagement metrics**
- **Uncovered gaps in country prioritization strategy** for Instagram ranking systems by identifying **16** countries currently classified as "Low Monetization Efficiency" that generate **6.4× the median revenue-per-capacity ROI** of their current classification
- **Built an interactive ML capacity tracking dashboard used by 10+ team members** to monitor real-time server usage and revenue metrics across countries and user demographics, backed by daily recurring **Presto data pipelines** aggregating metrics across **3B+ Instagram users**

SwingVision Berkeley, CA | January 2025 – May 2026

Data Science Intern

- **Deployed an AI tennis and pickleball highlights feature to 25K+ daily users** by training a **PyTorch neural network** and integrating it into the iOS app with **Swift/XCode**, enabling instant surfacing of the **top 5** points from **10K+** matches daily and enhancing user engagement
- **Automated social media content curation** from user generated content (UGC), driving **5M+ Instagram views, 1.5M+ TikTok views, 500K+ YouTube views, and 40K+ new followers** by analyzing **10M+** tennis points weekly to identify the most engaging highlights
- **Reduced marketing team content sourcing time by 80%+** by connecting to the **Google Sheets API** and implementing custom SQL queries tailored to different content niches, giving the marketing team access to high-engagement highlights for social media campaigns
- **Cut SQL query time by 90%+ across multiple data team projects** by securely connecting to the **AWS** database and converting SQL queries from Beekeeper Studio to **SQLAlchemy in Python**, removing the 50K-row limit and enabling in-script querying of **100M+** records
- **Saved data labelers 100+ labeling hours** by implementing a decision tree classifier to **filter 90%+ of irrelevant data**, leveraging tennis court location coordinates to achieve **100% recall** to curate a dataset to train a new computer vision model for a specific recording angle
- **Worked cross-functionally with business teams** to formulate creative data-driven approaches to **new product and marketing initiatives** such as personalized annual recaps and user improvement stats, bringing an analytical perspective into business strategy

HERO2 Berkeley, CA | August 2025 – December 2025

Machine Learning Researcher

- **Built a robust data-processing pipeline** that pulls jobs from **Redis** and retrieves user-generated phone sensor data from **MinIO**, integrating accelerometer, gyroscope, and GPS data for transportation classification, and delivers calculated CO_2 emissions offset to the backend
- **Designed trip-inference models using Random Forest decision trees** to classify transport modes and compute user CO_2-emission offsets, optimizing phone sensor collection frequencies to balance accuracy with battery usage, enabling dependable carbon-credit verification

PROJECTS

AI Fantasy Sports Assistant

- **Developed an LLM-powered fantasy sports assistant** utilizing **RAG** and the **Gemini API** to process 1,000+ real-time NFL, NBA, and MLB subreddit posts webscraped via the **Reddit API** to summarize player trends and provide actionable advice

NFL Passing Yards Projection Model

- **Built an XGBoost regressor to forecast weekly quarterback passing yards**, powered by a real-time data pipeline using the **NFL API** that integrated Next Gen Stats and Vegas betting lines, and achieved a **100%+ profitability** by exploiting player prop market inefficiencies

SKILLS

Languages: Python, SQL, Java, R, Swift, Bash, HTML

Technologies: Pandas, NumPy, SciPy, sci-kit learn, XGBoost, PyTorch, PySpark, Presto, Keras, TensorBoard, Scrapy, BeautifulSoup, spaCy, TextBlob, AWS, MongoDB, Docker, Redis, Git, Jira, Tableau, Matplotlib, Seaborn, Plotly, Folium, Claude Code, Cursor, Gemini API

Awards & Involvement: 2× UC Berkeley Datathon For Social Good Award Winner, 3× Dean's List, STEM Tutor at Berkeley High School